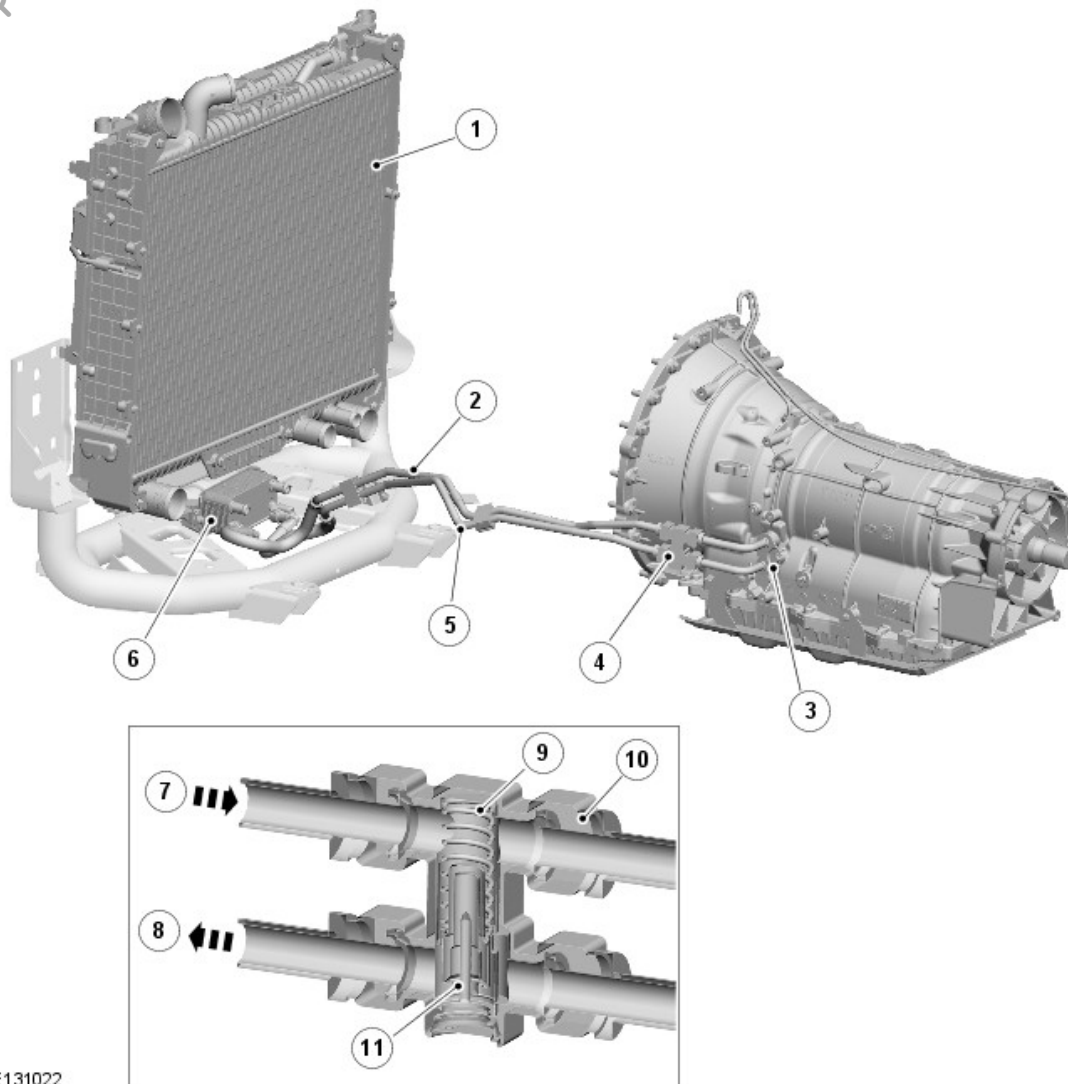


2016.0 RANGE ROVER (LG), 307-02

TRANSMISSION/TRANSAXLE COOLING - VEHICLES WITH: 8HP70 8-SPEED AUTOMATIC TRANSMISSION AWD

DESCRIPTION AND OPERATION

4.4L TDV8 FLUID COOLING COMPONENT
LOCATION



E131022

ITEM	DESCRIPTION
1	Engine cooling radiator
2	Return pipe (To transmission)
3	Transmission cooling pipe connection
4	Thermostatic valve
5	Supply pipe (From transmission)
6	Transmission fluid cooler
7	Fluid return flow
8	Fluid supply flow
9	Spring
10	Thermostatic valve body

INTRODUCTION

The transmission fluid cooler is a separate cooling unit located behind the engine cooling radiator. The unit is attached, via a bracket, to the cooling pack protection subframe. The transmission is connected to the fluid cooler via flexible hoses and metal pipes.

The transmission fluid cooler is a dedicated cooler which is connected into the engine cooling system circuit. The transmission fluid is cooled by the temperature differential between the transmission fluid and the engine coolant flowing through the cooler.

Fluid is supplied from the transmission fluid pump, through the thermostatic valve (if open) into the cooler. After passing through the cooler, the fluid passes out of the cooler and is returned to the transmission fluid pan.

The thermostatic valve is located in the transmission cooler fluid supply and return lines. The valve contains a bi-metallic spring and valve which is closed when transmission fluid temperatures are below 69 °C (156.2 °F). At temperatures of 69 °C (156.2 °F) and above, the valve begins to open allowing the fluid to flow around the cooler. The valve is fully open at temperatures of 79 °C (174.2 °F)